

723. CLINICAL ALLOGENEIC AND AUTOLOGOUS TRANSPLANTATION:
LATE COMPLICATIONS AND APPROACHES TO DISEASE RECURRENCE:
LATE COMPLICATIONS AND IMMUNE EVASION | DECEMBER 03, 2015

The Abundance of Certain Bacteria in the Intestinal Flora Is Associated with Relapse after Allogeneic Hematopoietic Stem Cell Transplantation

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Blood (2015) 126 (23) : 744.

<http://doi.org/10.1182/blood.V126.23.744.744>

Abstract



The major causes of mortality after allogeneic hematopoietic stem cell transplantation (allo-HSCT) are relapse, graft-versus-host disease (GVHD), and infection. We have previously reported that changes in the intestinal flora can affect GVHD, bacteremia, and overall survival. As intestinal bacteria are potent modulators of systemic immune responses, and since GVHD is correlated with graft-versus-tumor activity, we hypothesized that components of the intestinal flora could be associated with relapse after allo-HSCT.

We applied a biomarker-discovery approach and performed a retrospective observational analysis of 160 adults who received an unmodified (T-cell-replete) allograft. Patients were prospectively enrolled in a fecal biospecimen-collection protocol. For this analysis, we selected patients who had at least one specimen during the first 3 weeks following allo-HSCT. The primary diseases in this cohort were AML (37%), Non-Hodgkin's Lymphoma (33%), ALL (8%), MDS (7%), CLL (6%), Hodgkin's Lymphoma (6%),

CML (2%), and myeloproliferative neoplasm (2%). The mean age of the patients was 52 years (range 21-75). They were conditioned with ablative (17%), reduced-intensity (64%), and nonmyeloablative (19%) regimens. They received grafts from cord blood (46%), unrelated adults (33%), or related adults (22%). Among adult grafts, 92% were from peripheral blood and 8% were from bone marrow.

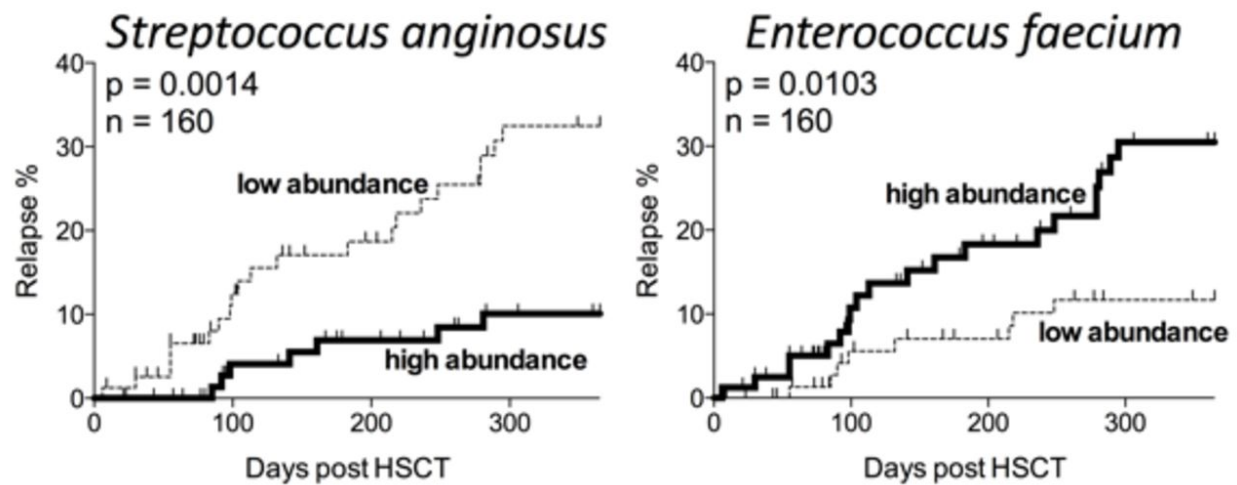
A census of the bacterial species in each stool sample was generated by 16S rRNA deep-sequencing as previously described (Jenq *et al.*, *BiolBone Marrow Transplant* 2015). The area under the curve of bacterial abundance over time was used as a measure of each patient's cumulative exposure to each bacterial taxon. Bacterial taxa of each patient present at a frequency >1% were evaluated for association with the outcome of relapse or progression of disease within the first year after allo-HSCT using linear discriminant analysis of effect size (LEfSe), a common approach in microbiota studies (Segata *et al.*, *Genome Biology*, 2011).

Among the taxons most significantly associated with freedom from relapse were members of the human oral flora including *Streptococcus anginosus*. After stratifying the patients by median abundance, we found that those with higher abundance of this bacterium had less relapse after transplantation (*Left figure*, $p = 0.0014$). We also identified bacteria associated with increased risk of relapse, such as *Enterococcus faecium* (*Right figure*, $p = 0.0103$).

We evaluated these bacteria as biomarkers in multivariate Cox models adjusted for three factors that were associated with relapse in this cohort: Refined Disease Risk Index (RDRI, Armand *et al.*, *Blood* 2014), conditioning intensity, and graft source (cord blood vs. adult donor). *Streptococcus anginosus* predicted relapse in a multivariate model adjusted for all three factors (HR 0.39, 95% CI 0.16-0.96, $p = 0.041$). *Enterococcus faecium* predicted relapse in a model adjusted for RDRI and conditioning intensity but failed to do so in a model additionally adjusted for graft source. In this analysis there was no formal adjustment for multiple comparisons; these data are now being validated in an additional cohort of patients whose samples are being sequenced. Finally, although we have previously reported that low bacterial diversity is associated with decreased overall survival after allo-HSCT (Taur *et al.*, *Blood* 2014), we did not find an association between bacterial diversity and relapse as assessed by reciprocal Simpson diversity index ($p > 0.1$).

Thus, the results of this retrospective analysis have identified an association between relapse after allo-HSCT and the abundance of two bacteria in the intestinal flora. These might serve as potential novel diagnostics or therapeutic targets to prevent relapse and improve overall survival after allo-HSCT.

Figure 1.



Disclosures

Peled: Merck: Research Funding. **Giralt:** SANOFI: Consultancy, Honoraria, Research Funding; TAKEDA: Consultancy, Honoraria, Research Funding; AMGEN: Consultancy, Research Funding; JAZZ: Consultancy, Honoraria, Research Funding, Speakers Bureau; CELGENE: Consultancy, Honoraria, Research Funding. **Perales:** Merck: Honoraria; Takeda: Honoraria; Amgen: Honoraria; Astellas: Honoraria; NMDP: Membership on an entity's Board of Directors or advisory committees. **van den Brink:** Boehringer Ingelheim: Consultancy, Other: Advisory board attendee; Novartis: Consultancy, Membership on an entity's Board of Directors or advisory committees; Tobira Therapeutics: Other: Advisory board attendee; Regeneron: Honoraria; Merck: Honoraria.

Author notes

*Asterisk with author names denotes non-ASH members.



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